Chapter 7. Regression Estimation

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1 The Regression Estimator

The ratio estimator works best when y is proportional to x (a line through the origin). If the relation is linear with a nonzero intercept, use the regression line instead. As in Ch. 6 the x's are treated as fixed (already observed) and the y's as random:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \beta_0' + \beta_1(x_i - \bar{x}) + \varepsilon_i, \qquad \hat{\beta}_0' = \bar{y}, \quad \hat{\beta}_1 = b = \frac{\sum(y_i - \bar{y})(x_i - \bar{x})}{\sum(x_i - \bar{x})^2},$$

so the fitted mean response is $\widehat{\mathrm{E}}(y \mid x) = \bar{y} + b(x - \bar{x})$. Evaluating it at the known population mean μ_x gives the estimator.

Definition 7.1

The **regression estimator** of μ_y is

$$\hat{\mu}_{yL} = \widehat{\mathrm{E}}(y \mid \mu_x) = \bar{y} + b(\mu_x - \bar{x}) = \frac{1}{n}\sum_{i=1}^{n}\left[y_i - b(x_i - \mu_x)\right],$$

i.e. the sample mean of the adjusted values $y_i - b(x_i - \mu_x)$.

If the sample happens to have $\bar{x} < \mu_x$ and $b > 0$, the sample mean $\bar{y}$ is probably too small as well, and $\hat{\mu}_{yL}$ corrects it upward by $b(\mu_x - \bar{x})$.

Theorem 7.2: Model unbiasedness

Under the model $\mathrm{E}(y_i) = \beta_0 + \beta_1 x_i$ with fixed x's,

$$\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \mu_x = \mathrm{E}(y \mid \mu_x).$$

Proof. $\mathrm{E}(\bar{y}) = \beta_0 + \beta_1 \bar{x}$ and $\mathrm{E}(b) = \beta_1$ (Ch. 6), so $\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \bar{x} + \beta_1(\mu_x - \bar{x}) = \beta_0 + \beta_1 \mu_x$. ■

2 Variance of the Regression Estimator

2.1 Fixed slope

First take a fixed constant β and $\bar{y}_{\text{reg}} = \bar{y} + \beta(\mu_x - \bar{x})$. This is the SRS mean of $u_i = y_i - \beta(x_i - \mu_x)$, whose population mean is μ_y, so it is exactly unbiased.

Theorem 7.3

$$\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n}(S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2),$$

which is minimized at $\beta = S_{xy}/S_x^2$ (the population LS slope), where

$$\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n}\Big(S_y^2 - \frac{S_{xy}^2}{S_x^2}\Big) = \frac{1-f}{n} S_y^2(1-\rho^2).$$

Proof. By Ch. 2, $\operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n} S_u^2$ with

$$\begin{aligned} S_u^2 &= \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i-\mu_y) - \beta(x_i-\mu_x)\big]^2 \\ &= \frac{1}{N-1}\Big[\sum(y_i-\mu_y)^2 - 2\beta\sum(x_i-\mu_x)(y_i-\mu_y) + \beta^2\sum(x_i-\mu_x)^2\Big] = S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2. \end{aligned}$$

This quadratic in β is minimized at $\beta = S_{xy}/S_x^2$, giving $S_y^2 - S_{xy}^2/S_x^2 = S_y^2(1-\rho^2)$. ∎

2.2 Estimated slope

With b estimated, write $\xi_i = (y_i - \mu_y) - \beta(x_i - \mu_x)$ for $\beta = S_{xy}/S_x^2$, so $y_i = \mu_y + \beta(x_i - \mu_x) + \xi_i$. Then

$$b = \frac{\sum(x_i-\bar{x})y_i}{\sum(x_i-\bar{x})^2} = \beta\frac{\sum(x_i-\bar{x})(x_i-\mu_x)}{\sum(x_i-\bar{x})^2} + \frac{\sum(x_i-\bar{x})\xi_i}{\sum(x_i-\bar{x})^2} = \beta + \frac{\sum(x_i-\bar{x})\xi_i}{\sum(x_i-\bar{x})^2},$$

and the last term is of smaller order since the ξ_i are uncorrelated with x in the population. Hence $b \approx \beta$ in large samples and

$$\operatorname{Var}(\hat{\mu}_{yL}) \approx \operatorname{Var}(\bar{y}_{\text{reg}}) = \frac{1-f}{n} S^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i-\mu_y) - \beta(x_i-\mu_x)\big]^2.$$

Theorem 7.4: Estimated variance

$$\widehat{\operatorname{Var}}(\hat{\mu}_{yL}) = \frac{N-n}{Nn} MS_E, \qquad MS_E = \frac{1}{n-2}\sum_{i=1}^{n}\big[(y_i-\bar{y}) - b(x_i-\bar{x})\big]^2,$$

the mean square error from the simple linear regression of y on x.

The sample analogue of S^2 is $\frac{1}{n-1}\sum e_i^2$ with residuals $e_i = (y_i - \bar{y}) - b(x_i - \bar{x})$; dividing by $n-2$ instead (two estimated coefficients) gives MS_E, and the difference is negligible for moderate n.

Example 7.5: Calculus marks

A mathematics achievement test was taken by all $N = 486$ entering students, with mean $\mu_x = 52$. For $n = 10$ sampled students:

Student	1	2	3	4	5	6	7	8	9	10
Test (x)	39	43	21	64	57	47	28	75	34	52
Final (y)	65	78	52	82	92	89	73	98	56	75

Here $\bar{x} = 46$, $\bar{y} = 76$, and the R fit gives $\hat{y} = 40.78 + 0.7656x$ with residual standard error 8.704 on 8 df ($MS_E = 75.76$, $R^2 = 0.705$). Then

$$\hat{\mu}_{yL} = 76 + 0.7656(52 - 46) = 80.6,$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{486 - 10}{486 \times 10}(75.76) = 7.42, \qquad B = 2\sqrt{7.42} = 5.45.$$

The fitted intercept (40.8) is far from zero, so a ratio estimator would be inappropriate here: $r\mu_x = (760/460)(52) = 85.9$ overshoots.

3 Relative Efficiency: Mean, Ratio and Regression

For two estimators E_1, E_2 of μ_y, $\mathrm{RE}(E_1/E_2) = \mathrm{Var}(E_2)/\mathrm{Var}(E_1)$; if it exceeds 1, E_1 is preferred. The estimated version (with estimated variances) is meaningful for large samples. The three estimated variances share the factor $\frac{1-f}{n}$:

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{1-f}{n}s_y^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_y) = \frac{1-f}{n}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y), \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{1-f}{n}s_y^2(1-\hat{\rho}^2).$$

Derivation: the regression variance in correlation form

$$\begin{aligned}\frac{1}{n-1}\sum\left[(y_i - \bar{y}) - b(x_i - \bar{x})\right]^2 &= \frac{1}{n-1}\left[S_{yy} - 2b\,S_{xy} + b^2 S_{xx}\right] \\ &= s_y^2 - 2b\,s_{xy} + b^2 s_x^2 = s_y^2 - b^2 s_x^2,\end{aligned}$$

since $s_{xy} = b s_x^2$. Moreover $\hat{\rho} = \frac{s_{xy}}{s_x s_y} = \frac{b s_x^2}{s_x s_y}$ gives $\hat{\rho} s_y = b s_x$, so $s_y^2 - b^2 s_x^2 = s_y^2(1-\hat{\rho}^2)$.

3.1 Mean versus ratio

From Ch. 5,

$$\widehat{\mathrm{RE}}(\hat{\mu}_y/\bar{y}) = \frac{s_y^2}{s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y},$$

and the ratio estimator is more efficient iff $\hat{\rho} > \frac{1}{2}\cdot\frac{s_x/\bar{x}}{s_y/\bar{y}}$. Recall that r is generally biased for R; the bias is negligible when the relation is a line through the origin.

3.2 Mean versus regression

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\bar{y}) = \frac{s_y^2}{s_y^2(1-\hat{\rho}^2)} = \frac{1}{1-\hat{\rho}^2} > 1 \quad \text{unless } \hat{\rho} = 0.$$

The regression estimator is (asymptotically) never worse than $\bar{y}$.

3.3 Ratio versus regression

Theorem 7.6

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\hat{\mu}_y) = \frac{s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y}{s_y^2(1-\hat{\rho}^2)} \geq 1,$$

with equality iff $r = b$.

Proof. The numerator minus the denominator is

$$r^2 s_x^2 - 2r\hat{\rho}s_x s_y + \hat{\rho}^2 s_y^2 = (rs_x - \hat{\rho}s_y)^2 = (rs_x - bs_x)^2 = s_x^2(r-b)^2 \geq 0.$$

■

Both r and b are linear in the y's: $r = \sum y_i / \sum x_i = \sum d_i y_i$ with equal weights $d_i = 1/\sum x_j$, while $b = \sum w_i y_i$ with $w_i = (x_i - \bar{x})/\sum (x_j - \bar{x})^2$. They coincide only when the LS line passes through the origin, which is exactly when the ratio estimator is as good as the regression estimator.